

A single 3-graph with infinite stability number

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Abstract

The stability number of a forbidden family measures how many different structures are needed to approximate all near-extremal constructions avoiding it. An infinite stability number means that no finite list of structures suffices. We construct a simple explicit 3-graph whose stability number is infinite. This extends the infinite-stability phenomenon for finite forbidden families, established by Hou–Li–Liu–Mubayi–Zhang, to the single-forbidden setting, and further develops the single-3-graph direction of Balogh–Clemen–Luo, in which exponentially many exact extremal constructions coexist with stability.

1 Introduction

The Turán problem goes back to Turán’s 1941 theorem [25], which determines the maximum number of edges in an n -vertex graph containing no copy of K_t and identifies the balanced complete $(t-1)$ -partite graph as the unique extremal construction. For $r \geq 2$, an r -uniform hypergraph, or r -graph, H is a collection of r -subsets of a finite set $V(H)$. We identify a hypergraph with its edge set, and hence write $|H|$ for the number of edges of H . Given a family $\mathcal{F}$ of r -graphs, the *Turán number* $\text{ex}(n, \mathcal{F})$ is the maximum number of edges in an $\mathcal{F}$ -free r -graph on n vertices, where $\mathcal{F}$ -free means containing no member of $\mathcal{F}$ as a subgraph. The *Turán density* is

$$\pi(\mathcal{F}) = \lim_{n \rightarrow \infty} \text{ex}(n, \mathcal{F}) / \binom{n}{r},$$

and if $\mathcal{F} = \{F\}$ we write $\pi(F)$.

For graphs, the Erdős–Stone–Simonovits theorem [8, 7] gives the asymptotic Turán density for every fixed forbidden graph, and the Simonovits stability theorem [24] shows that near-extremal graphs are close to the appropriate complete multipartite graph. Thus graph Turán problems are often governed by a small, rigid set of extremal constructions.

For hypergraphs, the situation is much less rigid; see Keevash’s survey [13]. The classical test case is Turán’s tetrahedron problem: determine the maximum size of a 3-graph containing no copy of the complete 3-graph K_4^3 . Turán conjectured the density to be $5/9$, but even the conjectural extremal picture is far from unique. Brown [4] and Kostochka [15] found further constructions for this problem, and Kostochka [15] proved that, if Turán’s conjecture is correct, then for $n = 3k$ there are at least 2^{k-2} non-isomorphic extremal hypergraphs; Frohmader [10] later constructed exponentially many conjectured extremal examples in the remaining congruence classes. This phenomenon suggests that hypergraph Turán problems should be studied not only through their densities, but also through the number and geometry of their extremal and near-extremal models.

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This paper works at the tripartite density $2/9$. A basic example at this density is the *generalized triangle*

$$F_5 = \{123, 124, 345\}.$$

Bollobás [3] proved the extremal theorem for cancellative triple systems, Frankl and Füredi [9] proved $\pi(F_5) = 2/9$ and the exact theorem for all sufficiently large n , and Keevash–Mubayi [14] gave stability refinements for cancellative hypergraphs. The same tripartite structure also appears in the enumeration theorem of Balogh and Mubayi [2], for almost all F_5 -free triple systems.

A *finite-template form of stability* was introduced by Mubayi [22] and, independently, by Pikhurko [23] to describe Turán problems in which all near-extremal examples can be approximated by one of finitely many model constructions.

Definition 1.1 (*stability number*). Let $\mathcal{F}$ be a family of r -graphs. We say that $\mathcal{F}$ is t -stable if, for every $m \in \mathbb{N}$, there exist r -graphs $G_1(m), \dots, G_t(m)$ on m vertices such that the following holds. For every $\delta > 0$ there exist $\epsilon > 0$ and N_0 such that, whenever $n \geq N_0$ and H is an n -vertex $\mathcal{F}$ -free r -graph with $|H| > (1 - \epsilon) \text{ex}(n, \mathcal{F})$, after relabelling, the hypergraph H can be transformed into some $G_i(n)$ by adding and deleting at most δn^r edges. The least such integer t is the *stability number* of $\mathcal{F}$ and is denoted by $\xi(\mathcal{F})$; if no such t exists, we put $\xi(\mathcal{F}) = \infty$.

After these formulations, Liu and Mubayi [18] constructed the first finite 3-graph family that is not stable in the ordinary one-model sense, and proved that it is nevertheless 2-stable. Liu, Mubayi and Reiher [19, 20] developed a unified stability framework and constructed, for every fixed t , finite families of triple systems with stability number t . Hou, Li, Liu, Mubayi and Zhang [12] then proved exact and stability results with infinitely many extremal constructions, using their crossed blowup operation. Related recent work includes the 2-stable family of Zhang, Hou and Li [28], the mixing-pattern constructions of Liu and Pikhurko [21], and the 1-stable examples of Balogh, Clemen and Luo [1] of single non-degenerate 3-graphs with exponentially many exact extremal constructions.

Although finite forbidden families are indispensable in this theory, the single-forbidden case is the basic unit of the classical Turán problem. The tetrahedron problem already points in this direction: Kostochka’s constructions [15] were used by Liu and Mubayi [18] to show that Turán’s conjecture $\pi(K_4^3) = 5/9$ would imply $\xi(K_4^3) = \infty$. Thus it is natural to ask whether this single-forbidden infinite-stability phenomenon can be proved unconditionally. Our main result answers this question affirmatively.

Let

$$K_4^- := \{abc, abd, acd\} \quad \text{and} \quad F^* := \{123, 124, 345, 156, 257\}.$$

Set

$$F := K_4^- \times F^*,$$

where the product is the categorical product defined in Section 2. Thus F has 28 vertices and $3 \cdot 5 \cdot 3! = 90$ edges.

Theorem 1.2. *The 3-graph F satisfies $\pi(F) = \frac{2}{9}$ and $\xi(F) = \infty$.*

The proof has three short components. First, we prove an auxiliary two-forbidden-graph theorem with density $2/9$. Second, a Brown–Simonovits blowup argument turns this auxiliary theorem into the upper bound for our single-forbidden graph. Finally, we use the one-parameter family of crossed blowups from [12]. In the language of Section 4, these crossed blowups are rank-two cut templates. We prove that every rank-two cut template is F -hom-free, while rank three already contains F homomorphically. Hence the crossed-blowup family produces asymptotically extremal F -free graphs that are pairwise far apart in edit distance, forcing $\xi(F) = \infty$.

2 Preliminaries

We first recall homomorphisms and the categorical product. A *homomorphism* $\phi : G_1 \rightarrow G_2$ between 3-graphs is a map $V(G_1) \rightarrow V(G_2)$ such that $\phi(e) \in G_2$ for every $e \in G_1$. We write $G_1 \rightarrow G_2$ if such a homomorphism exists and $G_1 \nrightarrow G_2$ otherwise. For a single 3-graph F , a 3-graph T is *F-hom-free* if $F \nrightarrow T$; equivalently, every blowup of T is *F-free*. For a family $\mathcal{F}$, we say that T is *F-hom-free* if $J \nrightarrow T$ for every $J \in \mathcal{F}$.

For 3-graphs G and H , their *categorical product* $G \times H$ is the 3-graph with vertex set

$$V(G \times H) = V(G) \times V(H),$$

where a triple $\{(x_1, y_1), (x_2, y_2), (x_3, y_3)\}$ is an edge if and only if

$$\{x_1, x_2, x_3\} \in G \quad \text{and} \quad \{y_1, y_2, y_3\} \in H.$$

In particular, the two coordinate projections are homomorphisms $G \times H \rightarrow G$ and $G \times H \rightarrow H$. This is the standard 3-uniform *direct product*. For $r = 2$, it is the categorical product of graphs, also called the direct, Kronecker or cardinal product, going back to Čulík [26] and Weichsel [27]. Hypergraph analogues were studied from a categorical viewpoint by Dörfler and Waller [6]; see also the survey of Hellmuth, Ostermeier and Stadler [11]. In the uniform Turán density setting, analogous product/colorability ideas appear in the palette approach: Lamaison [17] proved that palettes determine uniform Turán density, and Král', Kučerák, Lamaison and Tardos [16] use products and homomorphisms of palettes in their classification framework.

Let G be a 3-graph with vertex set $[m]$, and let $V_1, \dots, V_m$ be pairwise disjoint sets. The *blowup* $G[V_1, \dots, V_m]$ is obtained from G by replacing every vertex i by the set V_i and every edge ijk by the complete tripartite 3-graph with parts V_i, V_j, V_k . Equivalently, $J \rightarrow G$ if and only if J is contained in some blowup of G .

A family $\mathcal{F}$ of 3-graphs is *blowup-invariant* if every $\mathcal{F}$ -free 3-graph is $\mathcal{F}$ -hom-free.

The *Lagrange polynomial* of G is

$$p_G(x_1, \dots, x_m) = \sum_{ijk \in G} x_i x_j x_k.$$

If $|V_i| = x_i n + O(1)$ for all i , then the blowup $G[V_1, \dots, V_m]$ has

$$p_G(x)n^3 + O(n^2)$$

edges. Thus $p_G(x)$ records the normalized cubic edge count of the blowup with part proportions x , and the corresponding density normalized by $\binom{n}{3}$ is $6p_G(x) + o(1)$.

The *Lagrangian* of G is

$$\lambda(G) = \max\{p_G(x_1, \dots, x_m) : (x_1, \dots, x_m) \in \Delta_{m-1}\},$$

where Δ_{m-1} is the standard simplex. We also write

$$Z(G) = \{x \in \Delta_{m-1} : p_G(x) = \lambda(G)\}.$$

For two 3-graphs G and H with the same number of vertices, we write

$$\text{dist}(G, H) := \min_{\psi: V(G) \rightarrow V(H)} |\psi(G) \triangle H|$$

for their *edit distance*, where the minimum is over all bijections ψ . When the vertex set is fixed, this agrees with the usual edit distance after the best relabelling.

We will use the following standard consequence of the blowup theorem of Brown and Simonovits [5]. If $F \rightarrow G$, then adding G to a family that already contains F does not change the Turán density. More generally, if $F_i \rightarrow G_i$ for all $i \in [m]$ and $\mathcal{A}$ is any family of 3-graphs, then

$$\pi(\mathcal{A} \cup \{F_1, \dots, F_m\}) = \pi(\mathcal{A} \cup \{F_1, \dots, F_m, G_1, \dots, G_m\}).$$

We will also use *Zykov symmetrization*, originating with Zykov [29], in the blowup-invariant form used by Hou, Li, Liu, Mubayi and Zhang [12]: for a blowup-invariant family, an extremal hypergraph may be assumed *symmetrized*. Here symmetrized means that any two vertices not contained together in an edge are *twins*, i.e. they have the same link.

For a 3-graph H , its *shadow* is

$$\partial H = \{xy : \text{there exists } z \text{ with } xyz \in H\}.$$

We write $S_3(n)$ for the complete balanced tripartite 3-graph on n vertices. It has asymptotic edge density $2/9$.

3 The Turán density of F

We first record the lower bound. The product F is not tripartite. Indeed, the five vertices

$$(a, 1), \quad (d, 2), \quad (c, 3), \quad (b, 4), \quad (a, 5)$$

span the generalized triangle F_5 , with edges coming from $acd \times 123$, $abd \times 124$, and $abc \times 345$. Thus the complete balanced tripartite 3-graph $S_3(n)$ is F -free, and hence

$$\pi(F) \geq \frac{2}{9}.$$

For the upper bound, we show that the two small coordinate graphs K_4^- and F^* already determine a Turán problem with density $2/9$.

Lemma 3.1. $\pi(\{K_4^-, F^*\}) = \frac{2}{9}$.

Proof. The lower bound is given by $S_3(n)$. Indeed, K_4^- is not tripartite, and F^* contains the non-tripartite 3-graph $F_5 = \{123, 124, 345\}$. Hence $S_3(n)$ is $\{K_4^-, F^*\}$ -free and has density $2/9$.

It remains to prove the upper bound. Let $\mathcal{I}(F^*)$ be the finite family of all homomorphic images of F^* , and set

$$\mathcal{B} := \{K_4^-\} \cup \mathcal{I}(F^*).$$

Since $F^* \rightarrow J$ for every $J \in \mathcal{I}(F^*)$, repeated application of the Brown–Simonovits blowup theorem gives $\pi(\{K_4^-, F^*\}) = \pi(\{K_4^-\} \cup \mathcal{I}(F^*)) = \pi(\mathcal{B})$. The family $\mathcal{B}$ is *blowup-invariant*. Indeed, K_4^- is *2-covered*, meaning that every pair of vertices is contained in an edge, so every homomorphic image of K_4^- is K_4^- itself. Moreover, $\mathcal{I}(F^*)$ is closed under taking homomorphic images. Hence if a $\mathcal{B}$ -free 3-graph H admitted a homomorphism from some member of $\mathcal{B}$, then H would contain a member of $\mathcal{B}$, a contradiction.

By Zykov symmetrization, it suffices to consider symmetrized $\mathcal{B}$ -free 3-graphs. Let H be such a graph, and let T be its quotient obtained by taking one vertex from each equivalence class. Then H is a blowup of T . Moreover, the graph ∂T is complete: if two distinct quotient vertices were not contained together in any edge of T , then their representatives in H would be non-adjacent in the shadow and hence twins, contradicting that they lie in distinct quotient classes.

We claim that $F_5 \not\rightarrow T$. Suppose first that a homomorphism $F_5 \rightarrow T$ is not injective. Vertices lying in a common edge of F_5 cannot be identified, since the image of that edge must be a genuine 3-edge.

Since the only pairs of vertices of F_5 that are not contained together in an edge are $\{1, 5\}$ and $\{2, 5\}$, the only possible identifications are $5 = 1$ or $5 = 2$. In either case, the images of 123, 124, 345 form a copy of K_4^- , contradicting that T is K_4^- -free.

Thus every homomorphism $F_5 \rightarrow T$ would have to be injective. In this case T contains vertices, relabelled as 1, 2, 3, 4, 5, with edges 123, 124, 345. Since ∂T is complete, the pair 15 is contained in an edge 156 of T , and the pair 25 is contained in an edge 257 of T . The vertices 6 and 7 are allowed to coincide with previously named vertices, provided the displayed edges are genuine triples. Consequently, T contains a homomorphic image of F^* , contradicting that T is $\mathcal{I}(F^*)$ -free.

Therefore $F_5 \not\rightarrow T$. Since H is a blowup of T , the 3-graph H is F_5 -free. The Frankl–Füredi theorem [9] gives

$$|H| \leq \text{ex}(n, F_5) = \left(\frac{2}{9} + o(1)\right) \binom{n}{3}.$$

It follows that $\pi(\mathcal{B}) \leq 2/9$, completing the proof. $\square$

Corollary 3.2. $\pi(F) = \frac{2}{9}$.

Proof. The lower bound was proved at the beginning of the section. For the upper bound, the two coordinate projections give homomorphisms $F \rightarrow K_4^-$ and $F \rightarrow F^*$. By the Brown–Simonovits blowup theorem,

$$\pi(F) = \pi(\{F, K_4^-, F^*\}) \leq \pi(\{K_4^-, F^*\}) = \frac{2}{9},$$

where the last equality follows from Lemma 3.1. $\square$

4 Cut constructions and stability

We now present the lower-bound constructions used for stability. The key point is that the crossed blowups of Hou–Li–Liu–Mubayi–Zhang are exactly rank-two cut constructions. We prove that rank two is the largest cut-rank that is F -hom-free.

4.1 Cut templates

Let U be a finite-dimensional vector space over $\mathbb{F}_2$, and let $U^* = \text{Hom}_{\mathbb{F}_2}(U, \mathbb{F}_2)$ be its dual space. Let $\mathcal{C} \subseteq U^* \setminus \{0\}$ be a set of nonzero linear forms. The *cut template* $R(U, \mathcal{C})$ is the 3-graph with vertex set

$$\{a_c : c \in \mathcal{C}\} \sqcup U,$$

where

$$\{a_c, u, v\} \in R(U, \mathcal{C}) \iff c(u + v) = 1.$$

The vertices a_c are called *apex vertices*, the vertices of U are called *bottom vertices*, and the *cut-rank* is

$$\text{rank}(\mathcal{C}) := \dim_{\mathbb{F}_2} \langle \mathcal{C} \rangle.$$

Every edge of $R(U, \mathcal{C})$ contains exactly one apex vertex and two bottom vertices.

Equivalently, each form $c \in \mathcal{C}$ is a *graph cut* of the bottom set: since $c(u + v) = c(u) + c(v)$ over $\mathbb{F}_2$, the condition $c(u + v) = 1$ says precisely that u and v lie on opposite sides of the bipartition $U = c^{-1}(0) \sqcup c^{-1}(1)$. Thus the link graph of the apex a_c on the bottom vertices is the complete bipartite graph $K_{c^{-1}(0), c^{-1}(1)}$.

In ordinary graph-theoretic terms, $R(U, \mathcal{C})$ is obtained by placing one apex above each chosen cut and coning over all edges crossing that cut. The cut-rank is the $\mathbb{F}_2$ -rank of this family of cuts. In particular, the *full rank-two template* on $\mathbb{F}_2^2$ has four bottom vertices and three apex link graphs, corresponding to the three bipartitions of these four vertices into two pairs.

4.2 The rank dichotomy

We first record a small matrix observation.

Lemma 4.1. *Let M be a 3×3 zero-one matrix such that every permutation diagonal has sum one, i.e.*

$$M_{1,\sigma(1)} + M_{2,\sigma(2)} + M_{3,\sigma(3)} = 1 \quad \text{for all } \sigma \in S_3.$$

Then the three 1s of M lie either in a single row or in a single column.

Proof. Each entry of M lies on exactly two permutation diagonals. Summing the six diagonal equations gives $2 \sum_{i,j} M_{ij} = 6$, so M has exactly three 1s. If two 1s were in different rows and different columns, then some permutation diagonal would contain both, a contradiction. $\square$

Lemma 4.2. *Let M be a 4×3 zero-one matrix with rows indexed by a, b, c, d . Suppose that the three submatrices on rows abc , abd and acd all satisfy the condition in Lemma 4.1. Then one of the following holds:*

- (i) *the a -row is all 1s and the other three rows are all 0s, or*
- (ii) *one column is all 1s and the other two columns are all 0s.*

Proof. By Lemma 4.1, each of the three displayed 3×3 submatrices is of *row type* or *column type*. If one of them is of column type, then every submatrix sharing two of its rows must be of the same column type; otherwise the two common rows would receive incompatible patterns. Hence all three submatrices have the same full column of 1s.

It remains to consider the case where all three submatrices are of row type. The row of 1s must be common to the three submatrices. Indeed, if the row of 1s in the abc submatrix were, say, b or c , then the acd and abd submatrices would force two different non- a rows to be all 1s, contradicting row type in the remaining submatrix. Thus the common row is a . $\square$

Proposition 4.3. *If $\text{rank}(\mathcal{C}) \leq 2$, then $F \not\rightarrow R(U, \mathcal{C})$. Consequently, every blowup of $R(U, \mathcal{C})$ is F -free.*

Proof. It suffices to prove the statement for the full rank-two template

$$R_2 := R(\mathbb{F}_2^2, \{x, y, x + y\}),$$

where x, y are the two coordinate forms. Indeed, every cut template of rank at most two maps homomorphically to R_2 by recording the values of two basis cuts. Suppose, for a contradiction, that there is a homomorphism

$$\phi : F = K_4^- \times F^* \longrightarrow R_2.$$

Fix an edge $ijk \in F^*$. Define a 4×3 zero-one matrix by

$$M_{r,v} = 1 \iff \phi(r, v) \text{ is an apex vertex of } R_2, \quad r \in \{a, b, c, d\}, \quad v \in \{i, j, k\}.$$

For each of the three edges abc, abd, acd of K_4^- and for every bijection from this edge to $\{i, j, k\}$, the corresponding triple is an edge of F . Since every edge of R_2 contains exactly one apex vertex, the three submatrices on rows abc, abd and acd satisfy Lemma 4.1. Lemma 4.2 gives two possibilities for the edge ijk : either the a -row is the unique all-apex row, or one of the three columns is all-apex.

These two possibilities cannot be mixed on intersecting edges of F^* , because the column pattern of the common vertex would have to be both $(1, 0, 0, 0)^T$ and either $(1, 1, 1, 1)^T$ or $(0, 0, 0, 0)^T$. Since the edge-intersection graph of F^* is connected, all edges of F^* have the same type.

First suppose that all edges have column type. Then there is a zero-one labelling $s_1, \dots, s_7$ of $V(F^\star)$ such that every edge of $F^\star$ contains exactly one vertex with label 1. Hence

$$\begin{aligned} s_1 + s_2 + s_3 &= 1, & s_1 + s_2 + s_4 &= 1, & s_3 + s_4 + s_5 &= 1, \\ s_1 + s_5 + s_6 &= 1, & s_2 + s_5 + s_7 &= 1. \end{aligned}$$

The first two equations give $s_3 = s_4$. The third gives $s_3 = s_4 = 0$ and $s_5 = 1$. Then the fourth gives $s_1 = s_6 = 0$, the first gives $s_2 = 1$, and the fifth becomes $1 + 1 + s_7 = 1$, impossible.

Thus all edges have row type. For each $i \in V(F^\star)$ let $c_i \in \{x, y, x + y\}$ be the cut corresponding to the apex vertex $\phi(a, i)$. For $r \in \{b, c, d\}$, let

$$b_i^r \in \mathbb{F}_2^2$$

be the bottom point corresponding to $\phi(r, i)$. If $ijk \in F^\star$, then the edge condition first gives $c_i(b_j^p + b_k^q) = 1$ whenever $p \neq q$. To see that row differences of the same bottom vertex are annihilated by c_i , take distinct rows $p, p' \in \{b, c, d\}$ and choose q different from both. Applying the edge condition with bottom rows (p, q) and (p', q) gives

$$c_i(b_j^p + b_k^q) = c_i(b_j^{p'} + b_k^q) = 1,$$

and hence $c_i(b_j^p + b_j^{p'}) = 0$. The same argument applies to the k -vertex. If $p = q$, choose $p' \neq p$; then

$$c_i(b_j^p + b_k^p) = c_i(b_j^{p'} + b_k^p) + c_i(b_j^p + b_j^{p'}) = 1 + 0 = 1.$$

Consequently, for every two rows $p, q \in \{b, c, d\}$, possibly equal,

$$c_i(b_j^p + b_k^q) = 1, \quad c_i(b_j^p + b_j^q) = c_i(b_k^p + b_k^q) = 0. \quad (1)$$

Apply this to the two edges 123 and 124. For instance, applying (1) to 123 and 124 with apex 1 gives

$$c_1(b_2^r + b_3^p) = 1, \quad c_1(b_2^r + b_4^q) = 1,$$

and hence $c_1(b_3^p + b_4^q) = 0$. The same argument with apex 2 gives the corresponding c_2 -equation. Thus

$$c_1(b_3^p + b_4^q) = c_2(b_3^p + b_4^q) = 0 \quad \text{for all } p, q,$$

while the edge 345, from the apex vertex 5, gives

$$c_5(b_3^p + b_4^q) = 1 \quad \text{for all } p, q.$$

We now spell out the consequences of these equations. If $c_1 \neq c_2$, then $\ker c_1 \cap \ker c_2 = \{0\}$, so all vectors $b_3^p + b_4^q$ would be zero, contradicting $c_5(b_3^p + b_4^q) = 1$. Thus $c_1 = c_2$. Also $c_5 \neq c_1$. Since the system

$$c_1(v) = 0, \quad c_5(v) = 1$$

has a unique solution in $\mathbb{F}_2^2$, all vectors $b_3^p + b_4^q$ are equal. Hence the bottom points assigned to vertices 3 and 4 are independent of the row indices. Any invertible linear transformation of $\mathbb{F}_2^2$ permutes the three nonzero forms $x, y, x + y$ and induces an automorphism of R_2 , so after a linear change of coordinates we may assume

$$c_1 = c_2 = x.$$

The bottom points assigned to vertices 1 and 2 are also independent of the row indices. Indeed, row differences for vertex 1 lie in both $\ker c_2 = \ker x$ and $\ker c_5$, and row differences for vertex 2 lie in both $\ker c_1 = \ker x$ and $\ker c_5$; since $c_5 \neq x$, both intersections are zero. Translating all bottom points by a fixed vector preserves $u + v$ in characteristic two and hence induces an automorphism of R_2 , so we

may write $b_1 = (0, 0)$. The equations with apices 1 and 2 on the edge 123 give $x(b_1 + b_2) = 0$, while the equation with apex 3 gives $c_3(b_1 + b_2) = 1$; hence b_2 is the nonzero vector in $\ker x$. Thus

$$b_1 = (0, 0), \quad b_2 = (0, 1).$$

The equations from the apex 1 in the edges 123 and 124 give $x(b_3) = x(b_4) = 1$, and the edge 345 gives $b_3 \neq b_4$. Therefore

$$b_3 = (1, s), \quad b_4 = (1, s + 1)$$

for some $s \in \mathbb{F}_2$. Since $c_3(b_1 + b_2) = c_4(b_1 + b_2) = 1$, both c_3 and c_4 have the form

$$c_3 = p_3x + y, \quad c_4 = p_4x + y$$

with $p_3, p_4 \in \mathbb{F}_2$.

Write $b_5^r = (u_r, v_r)$ for $r \in \{b, c, d\}$. From the edge 345 we obtain, for every row r ,

$$p_3(1 + u_r) + s + v_r = 0, \quad p_4(1 + u_r) + s + v_r = 1.$$

Adding gives $(p_3 + p_4)(1 + u_r) = 1$, and hence $u_r = 0$ for every r .

Finally, the edges 156 and 257 give

$$c_6(b_1 + b_5^r) = 1, \quad c_7(b_2 + b_5^r) = 1 \quad \text{for every } r.$$

But $b_1 = (0, 0)$, $b_2 = (0, 1)$, and $b_5^r = (0, v_r)$. If $v_r = 0$, then the first equation evaluates a linear form on the zero vector; if $v_r = 1$, then the second equation does. Both are impossible. $\square$

Proposition 4.4. *If $\text{rank}(\mathcal{C}) \geq 3$, then $F \rightarrow R(U, \mathcal{C})$. Thus Proposition 4.3 is sharp.*

Proof. Choose three independent cuts $c^1, c^2, c^3 \in \mathcal{C}$. Let X, Y, Z be the coordinate forms on $\mathbb{F}_2^3$. The linear map

$$\pi : U \rightarrow \mathbb{F}_2^3, \quad \pi(u) = (c^1(u), c^2(u), c^3(u))$$

is surjective, so it has a linear right inverse $s : \mathbb{F}_2^3 \rightarrow U$. Therefore the map sending a bottom vertex $w \in \mathbb{F}_2^3$ to $s(w)$ and the apex vertices a_X, a_Y, a_Z to $a_{c^1}, a_{c^2}, a_{c^3}$, respectively, is a homomorphism

$$R(\mathbb{F}_2^3, \{X, Y, Z\}) \rightarrow R(U, \mathcal{C}).$$

Thus it is enough to give a homomorphism into $R(\mathbb{F}_2^3, \{X, Y, Z\})$.

Set

$$d_1 = d_2 = d_6 = d_7 = X, \quad d_3 = d_4 = Y, \quad d_5 = Z,$$

and choose bottom points

$$\begin{aligned} u_1 = 000, \quad u_2 = 010, \quad u_3 = 100, \quad u_4 = 101, \\ u_5 = 110, \quad u_6 = 001, \quad u_7 = 001. \end{aligned}$$

For every edge $ijk \in F^*$, the required evaluations are

ijk	$d_i(u_j + u_k)$	$d_j(u_i + u_k)$	$d_k(u_i + u_j)$
123	1	1	1
124	1	1	1
345	1	1	1
156	1	1	1
257	1	1	1

and hence

$$d_i(u_j + u_k) = d_j(u_i + u_k) = d_k(u_i + u_j) = 1.$$

Hence the map

$$(a, i) \mapsto a_{d_i}, \quad (r, i) \mapsto u_i \quad (r \in \{b, c, d\})$$

is a homomorphism from $K_4^- \times F^*$ to $R(\mathbb{F}_2^3, \{X, Y, Z\})$, and hence, by the lifting observation above, to $R(U, \mathcal{C})$. $\square$

4.3 Crossed blowups and infinite stability

Let

$$R_\times := R(\mathbb{F}_2^2, \{x, y\}).$$

This is the rank-two cut-template version of the *crossed-blowup* construction of Hou, Li, Liu, Mubayi and Zhang [12]. The following elementary construction records the special case needed here.

Theorem 4.5. *There is a family of $R_\times$ -blowups $G_\alpha(n)$, indexed by $\alpha \in (0, 1/2)$, such that*

$$|G_\alpha(n)| = \left(\frac{1}{27} + o(1)\right) n^3 = \left(\frac{2}{9} + o(1)\right) \binom{n}{3},$$

and for every distinct $\alpha, \beta \in (0, 1/2)$,

$$\text{dist}(G_\alpha(n), G_\beta(n)) = \Omega_{\alpha, \beta}(n^3).$$

Proof. Let the two apex classes corresponding to a_x and a_y be A_x and A_y , and let the four bottom classes corresponding to $00, 01, 10, 11 \in \mathbb{F}_2^2$ be $B_{00}, B_{01}, B_{10}, B_{11}$. Choose their sizes so that they sum to n and satisfy

$$|A_x| = |A_y| = \frac{n}{6} + O(1), \quad |B_{00}| = |B_{11}| = \frac{\alpha n}{3} + O(1), \quad |B_{01}| = |B_{10}| = \frac{(1-\alpha)n}{3} + O(1).$$

Let $G_\alpha(n)$ be the corresponding blowup of $R_\times$. Each side of each of the two cuts on the bottom vertices has size $n/3 + O(1)$, and hence

$$|G_\alpha(n)| = |A_x|(|B_{00}| + |B_{01}|)(|B_{10}| + |B_{11}|) + |A_y|(|B_{00}| + |B_{10}|)(|B_{01}| + |B_{11}|) = \frac{n^3}{27} + O(n^2).$$

It remains to prove separation. For a 3-graph H , write

$$Q(H) := \sum_{\{u, v\} \in \binom{V(H)}{2}} d_H(u, v)^2,$$

where $d_H(u, v)$ is the codegree of u and v . If two n -vertex 3-graphs differ in k edges, then changing these edges changes Q by at most $3(2n+1)k$, since changing one edge changes the codegrees of exactly three pairs by 1.

For $G_\alpha(n)$, the apex-bottom pairs contribute $2n^4/81 + o(n^4)$ to Q . Among bottom pairs, pairs differing in exactly one coordinate contribute $\alpha(1-\alpha)n^4/81 + o(n^4)$, while pairs differing in both coordinates contribute $(\alpha^2 + (1-\alpha)^2)n^4/81 + o(n^4)$. Thus

$$Q(G_\alpha(n)) = \left(\frac{3-\alpha+\alpha^2}{81} + o(1)\right) n^4.$$

The function $3-\alpha+\alpha^2$ is strictly decreasing on $(0, 1/2)$. Hence, for distinct $\alpha, \beta \in (0, 1/2)$, the quantities $Q(G_\alpha(n))$ and $Q(G_\beta(n))$ differ by $\Omega_{\alpha, \beta}(n^4)$. Since Q is invariant under relabelling, the preceding Lipschitz bound implies $\text{dist}(G_\alpha(n), G_\beta(n)) = \Omega_{\alpha, \beta}(n^3)$. $\square$

Proof of Theorem 1.2. Corollary 3.2 gives $\pi(F) = 2/9$. It remains to prove infinite stability. By Proposition 4.3, every $G_\alpha(n)$ in Theorem 4.5 is F -free, and by Corollary 3.2 these graphs are asymptotically extremal for F .

Suppose, for a contradiction, that $\xi(F) \leq t$. Choose distinct parameters $\alpha_1, \dots, \alpha_{t+1} \in (0, 1/2)$. By Theorem 4.5, the finitely many pairwise separation constants have a positive minimum $c > 0$. Thus, for all sufficiently large n ,

$$\text{dist}(G_{\alpha_i}(n), G_{\alpha_j}(n)) \geq cn^3 \quad \text{for all } i \neq j.$$

Set $\delta = c/3$. By t -stability, fix the corresponding template graphs $H_1(m), \dots, H_t(m)$ for each m . For this value of δ , there are $\epsilon > 0$ and N_0 such that every n -vertex F -free graph with more than $(1 - \epsilon) \text{ex}(n, F)$ edges is within δn^3 edits of one of the $H_t(n)$. For all sufficiently large n , each $G_{\alpha_i}(n)$ satisfies this edge condition, since it is asymptotically extremal. By the pigeonhole principle, two of the $t + 1$ graphs $G_{\alpha_i}(n)$ and $G_{\alpha_j}(n)$ are both within δn^3 edits of the same template. Hence, by the triangle inequality for edit distance, their edit distance is at most $2\delta n^3 < cn^3$, a contradiction. Therefore $\xi(F) = \infty$. $\square$

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Declaration on the use of AI

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